(Please write your Enrolment No. Immediately) *Enrollment No:* .....

Mid-Term Examination – 2025

B. TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

Fourth Semester, March, 2025

Paper Code: BS-202 **Subject:** Probability, statistics, and linear prog

Time: 1½Hrs. **Max. Marks:** 30

Note: Attempt Q. No. 1 which is compulsory and any two more questions from the remaining.

Q. No.	Question	Marks	CO
1(a)	Find the binomial distribution for which mean = 2(variance) and mean + variance = 3, and find P ($X \leq 3$).	[2.5]	CO1
1(b)	A manufacturer produces medicine bottles and finds that 0.1% of bottles are defective. The bottles are packed in boxes containing 500 bottles. Find that in 100 such boxes are expected to contain no defective bottles.	[2.5]	CO1
1(c)	Find k so that $f(x,y) = kxy$, $1 \leq x \leq y \leq 2$ will be a joint probability density function.	[2.5]	CO2
1(d)	Calculate the covariance of the following pairs of observations of the variables X and Y (15,44), (20,43), (25,45), (30,37), (40,34), (50, 37).	[2.5]	CO2
2(a)	The probability density function of a variate X is X : 0 1 2 3 4 5 6 P(X): K 3K 5K 7K 9K 11K 13K Find (i) P(X<4) (ii) $P(3 < X \leq 6)$ (iii) What will be the minimum value of k so that $P(X \leq 2) > 3$.	[5]	CO1
2(b)	There are three bags: the First contains 1 white, 2 red, and 3 black balls, the second contains 2 white,3 red, and 1 black balls, and the third contains 3 white,1 red, and 2 black balls. Two balls are drawn from a bag chosen at random. These are found to be one white and one red. Find the probability that the balls so drawn came from the second bag.	[5]	CO1
3(a)	A sample of 100 dry battery cells was tested to find the length of life produced the following result, μ = 12 hours and variance = 3 hours. If the data is normally distributed, what percentage of battery cells are expected to have a life of (i) more than 15 hours (ii) less than 6 hours (iii) between 10 and 14 hours? (Given that P(0<z<1) = 0.3413, P(0<z<0.67) = 0.2487,$P(0 < z < 2) = 0.4772$)	[5]	CO1
3(b)	Find the coefficient of correlation for 6 college students who got the following percentage of marks in Economics and Statistics: Economics: 10, 14, 18, 22, 26, 30 Statistics: 18, 12, 24, 6, 30, 36	[5]	CO2
4(a)	**(a)** Three fair of coins are tossed. Let X denote the number of heads on the first two coins, Let Y denote the number of heads on the last two coins. Then (i) Find the joint distribution of X and Y. (ii) Find E[Y|X=1].	[5]	CO2
4(b)	**(b)** The probability density function of a continuous bivariate distribution is given by F (x, y) = $\begin{cases} x + y, \text{where } 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0, otherwise. \end{cases}$ (i) Find the marginal distributions(ii)Find the correlation coefficient of X and Y.	[5]	CO2